

VIJAI VENKATESH NATARAJAN GANESH BABU

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EDUCATION

University of Michigan

MSE in Aerospace Engineering (Structures & Materials) - GPA: 3.727

College of Engineering Guindy

BE Manufacturing Engineering - GPA: 3.71

Ann Arbor, MI

Aug. 2024 - May 2026

Chennai, India

Apr. 2024

EXPERIENCE

Graduate Research Assistant | Taub Group, University of Michigan

Jan. 2025 - Present

- Executed systematic process optimization of flax fiber surface treatments using supercritical fluid technology (CO_2 , N_2 , Ar) with TiO_2 nanoparticle functionalization to enhance mechanical performance; fabricated composite laminates directly from treated fibers and conducted comparative tensile testing to assess mechanical property improvements and structural performance between treated and untreated fiber systems.
- Developed automated data processing script for single flax fiber tensile analysis, significantly reducing analysis time and improving measurement consistency across multiple tests.

Project Management Consultant | NeoNest Global

Sep. 2025 - Dec. 2025

- Developed scalable manufacturing plan for medical device clinical trials by conducting supplier feasibility studies and shortlisting 10-15 NDA-compliant manufacturers with ISO 13485 certification across selected countries.
- Built comprehensive manufacturer evaluation framework assessing technical capabilities, quality systems, regulatory compliance, production capacity, and regional viability to ensure supply chain readiness for clinical trial production.

Intern | Hindustan Aeronautics Limited

July 2022 - August 2022

- Conducted GD&T analysis and dimensional inspection of 50+ aerospace components using coordinate measuring machine and precision metrology instruments including height gauges, bore gauges, and micrometers.
- Reviewed engineering drawings and inspection documentation to ensure measurement compliance with aerospace manufacturing quality standards and specifications.

PROJECTS

Design and Manufacturing of Composite Laminates for Flexural Performance

Aug. 2025 - Dec. 2025

- Designed and manufactured 12-ply IM7-8552 carbon/epoxy composite laminate using Classical Lamination Theory, vacuum bagging, and autoclave curing, achieving target flexural modulus of 108 GPa.
- Conducted ASTM D7264 four-point bending tests achieving 83.64 ± 5.62 GPa experimental flexural modulus and validated CLT predictions within 3.2% through systematic error analysis and machine compliance correction.

Surrogate Modeling of a Composite Laminated Plate using a Physics Based Model

Jan. 2025 - Apr. 2025

- Developed physics-based model using ComposiPy library to predict structural performance metrics including buckling load factor, bending deflection, and natural frequency of composite laminated plates, reducing dependence on computationally intensive FEM simulations.
- Implemented Morris Screening sensitivity analysis to identify critical design variables and applied Latin Hypercube Sampling (LHS) for design of experiments (DOE) to generate training datasets; developed a surrogate model to approximate FEM results, achieving significant reduction in computational time.

Analysis and Redesign of Gear Pump

Aug. 2024 - Dec. 2024

- Conducted FAST analysis and optimized assembly sequence, improving manufacturing efficiency by 15%.
- Developed and maintained Bills of Materials (BOMs) for redesigned components, ensuring compliance with manufacturing constraints and design requirements.
- Redesigned mechanical components using Fusion 360 and developed detailed engineering drawings with geometric dimensioning and tolerancing (GD&T) specifications; optimized assembly design and BOMs, improving manufacturability and assembly efficiency by 40% and reducing manufacturing costs by 15%.

Laser Texturing of Ti-6Al-4V Alloy for Dental Implant Applications

Jan. 2024 - May 2024

- Designed and fabricated bio-inspired CO_2 laser surface texturing patterns on Ti-6Al-4V titanium alloy substrates to modify surface roughness and wettability characteristics, achieving 44% improvement in hydrophobicity for enhanced osseointegration performance.
- Conducted materials characterization of laser-textured surfaces using scanning electron microscopy (SEM) and water contact-angle to establish correlations between surface morphology and mechanical and wettability properties.

Fabrication and Hybridization of Hybrid Ballistic Composites

Aug. 2023 - Jan. 2024

- Fabricated sustainable Kevlar-29/flax hybrid composite laminates using vacuum bagging process, integrating natural fibers to reduce environmental impact while maintaining impact resistance and energy absorption for ballistic applications.
- Conducted high-velocity impact testing using gas gun apparatus to evaluate ballistic performance and failure behavior of hybrid composite laminates.

Agri-Spraying Drone

Aug. 2022 – Apr. 2023

- Designed and prototyped autonomous multi-rotor agricultural drone with 4 kg payload capacity, including a concept for automated docking station.
- Led cross-functional 15-member team through design, fabrication, and assembly phases including CAD modeling of structural and motor mounting platforms in SolidWorks, machining of Al 6061-T6 aluminum components, and integration of fully functional prototype.

SKILLS

Software: SolidWorks | Ansys | MATLAB | Microsoft Office

Manufacturing: Advanced Manufacturing | Composite Manufacturing | DFMA | Rapid Prototyping | Lean Six Sigma | Process Planning & Cost Estimation

Machine Tools: Tensile Testing | Diamond Saw | Grinding | Polishing | Materials Characterization

Technical : Electrochemical Energy Storage Systems | Composite Design | Solid Mechanics